

ISI

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10/5
AB
B
Subharish JV
1999
81008246001
Test Code: MEI/MEII 1998

Syllabus for MEI

Matrix Algebra : Matrices and Vectors, Matrix Operations, Determinants, non-singularity, Inversion, Cramer's rule.

Calculus : Limits, Continuity, Differentiation of functions of one or more variables, Product rule, Partial and total derivatives, Derivatives of implicit functions, unconstrained optimisation (first and second order conditions for extrema of functions of single variable, and several variables), Taylor series, Definite and indefinite integrals standard formulae, integration by parts and integration by substitution, Differential equations.

Constrained optimisation of functions of a single variable.

Theory of sequences and series.

Linear programming : Formulation, Statements of primal and dual problems, Graphical solutions.

Theory of polynomial equations (up to third degree).

Sample Questions MEI (Mathematics) 1998-99

Each question carries 4 marks.

1. Consider the sets

$$S_1 = \{0\}, \quad S_2 = \left\{\frac{3}{2}, \frac{5}{2}\right\}, \quad S_3 = \left\{\frac{8}{3}, \frac{11}{3}, \frac{14}{3}\right\},$$

$$S_4 = \left\{\frac{15}{4}, \frac{19}{4}, \frac{23}{4}, \frac{27}{4}\right\}, \quad S_5 = \left\{\frac{24}{5}, \frac{29}{5}, \frac{34}{5}, \frac{39}{5}, \frac{44}{5}\right\}, \dots$$

Then the sum of the elements in S_n is

- (A) $\frac{(n+1)((n-2))}{2}$ (B) $\frac{(n-1)(n+2)}{2}$
(C) $\frac{(n+1)(3n-2)}{2}$ (D) $\frac{(n-1)(3n+2)}{2}$

2. The value of $\log_2 e - \log_4 e + \log_8 e - \log_{16} e + \dots$ is equal to

- (A) $\frac{1}{2} \log_2 e$ (B) $\frac{1}{2} \log_2 2$ (C) $\frac{1}{2}$ (D) none of these.

[1]

3. $-\frac{1}{\alpha}$ and $-\frac{1}{\beta}$ are the roots of the quadratic equation $3x^2 - 18x + 2 = 0$.

The equation whose roots are α and β is

- (A) $2x^2 + 18x + 3 = 0$ (B) $2x^2 - 18x + 3 = 0$
(C) $3x^2 + 27x + 2 = 0$ (D) $3x^2 - 27x + 2 = 0$

4. The values of (a, b) for which the following system of equations

$$ax + 3y + z = 5$$

$$2x - y + 2az = 3$$

$$x + 4y + az = b$$

is inconsistent, are

- (A) $(a \neq 1, b = 6)$ and $(a \neq -1, b \neq \frac{66}{5})$
(B) $(a = 1, b \neq 6)$ and $(a \neq -1, b \neq \frac{66}{5})$
(C) $(a \neq 1, b = 6)$ and $(a \neq -1, b \neq \frac{66}{5})$
(D) $(a = 1, b \neq 6)$ and $(a = -1, b \neq \frac{66}{5})$

5. Maximum value of $f(x) = \frac{a^n}{n!}$, for $n = 0, 1, 2, 3, \dots$ where a is a given positive integer, is

- (A) a^a (B) $\frac{a^{a-1}}{a!}$ (C) $\frac{a^{a-1}}{(a-1)!}$ (D) none of these.

6. If $y = \log_{g(x)} f(x)$ where $f(x) = e^x$, $g(x) = x^2$, defined for $x > 1$, then $\frac{dy}{dx}$ is

- (A) $\frac{\log_e x - 1}{2(\log_e x)^2}$ (B) $\frac{\log_e x - 1}{(\log_e x)^2}$
(C) $\frac{\log_e x + 1}{2(\log_e x)^2}$ (D) $\frac{\log_e x + 1}{(\log_e x)^2}$

7. If $f(x) = \begin{cases} 3x^2 + 12x - 1, & -1 \leq x < 2 \\ 37 - x, & 2 \leq x \leq 3, \end{cases}$

then

- (A) $f(x)$ is not continuous in $[-1, 3]$
- (B) $f(x)$ is continuous and differentiable in $[-1, 3]$
- (C) $f(x)$ is continuous in $[-1, 3]$, but $f'(2)$ does not exist
- (D) none of these.

8. The following matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & -1 \\ 4 & 1 & 0 & 1 \\ 8 & 1 & 0 & -1 \end{bmatrix}$$

- (A) is singular and the rank is 3
- (B) is singular and the rank is less than 3
- (C) is nonsingular and the determinant is -12
- (D) none of these.

9. The roots of the cubic equation $x^3 - 9x^2 + 23x - 15 = 0$ are in AP. The common difference between the roots is

- (A) $1/2$
- (B) 2
- (C) 3
- (D) none of these.

10. If $\alpha_1, \alpha_2, \alpha_3, \dots$ are in GP, then the value of

$$\Delta = \begin{vmatrix} \log \alpha_n & \log \alpha_{n+1} & \log \alpha_{n+2} \\ \log \alpha_{n+3} & \log \alpha_{n+4} & \log \alpha_{n+5} \\ \log \alpha_{n+6} & \log \alpha_{n+7} & \log \alpha_{n+8} \end{vmatrix}$$

is

- (A) $\log (\alpha_n \alpha_{n+8}) - 2 \log \alpha_{n+4}$
- (B) $\log (\alpha_1 \alpha_3) - 2 \log \alpha_2$
- (C) 0
- (D) 1

[3]

11. The function $f(x)$ defined by

$$f(x) = \begin{cases} x & \text{when } 0 < x < 1 \\ 2-x & \text{when } 1 \leq x \leq 2 \\ x - \frac{1}{2}x^2 & \text{when } x > 2, \end{cases}$$

is

- (A) continuous at $x = 2$ but not differentiable at $x = 1$
- (B) continuous at $x = 2$ and differentiable at $x = 1$
- (C) not continuous at $x = 2$ and not differentiable at $x = 1$
- (D) none of these.

12. $\lim_{x \rightarrow 0} \left[\frac{1}{x} - \frac{\log(1+x)}{x^2} \right]$ is

- (A) 1 (B) $1/2$ (C) does not exist (D) none of these.

13. If $|a| < 1$, $|b| < 1$, then the series

$$a(a+b) + a^2(a^2+b^2) + a^3(a^3+b^3) + \dots$$

- (A) converges to $\{a^2/(1-a^2) + b^2/(1-b^2)\}$
- (B) converges to $\{a^2/(1-a^2) + b^2/(1-ab)\}$
- (C) converges to $\{a^2/(1-a^2) + ab/(1-ab)\}$
- (D) none of the above is true.

14. if $a + \frac{1}{a} = 2 \cos \alpha$, then the value of $a^n + \frac{1}{a^n}$ is

- (A) $2^n \cos^n \alpha$ (B) $2^n \cos n\alpha$ (C) $2 \cos n\alpha$ (D) none of these.

15. If a, b and c are positive real numbers, then the least value of

$$(a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \text{ is}$$

- (A) 9 (B) 3 (C) $\frac{10}{3}$ (D) none of these.

16. If

$$f(x) = \begin{cases} |x| - [x] - 1 & \text{if } -1 < x < 0 \\ |x| - [x] & \text{if } 0 \leq x < 1 \\ 1 & \text{otherwise,} \end{cases}$$

where $[x]$ is the greatest integer $\leq x$ and $|x|$ is the absolute value of x , then

- (A) $f(x)$ is a discontinuous function with points of discontinuity at only one point on the real line
 - (B) $f(x)$ is a discontinuous function with points of discontinuity at more than one point on the real line
 - (C) $f(x)$ is a continuous function but not differentiable at only one point on the real line
 - (D) none of these.
17. If a primal linear programming problem has feasible solutions, but no optimal solution then its dual problem
- (A) may have an optimal solution
 - (B) has no feasible solution
 - (C) has feasible solutions, but no optimal solution
 - (D) cannot be formulated.

18. Consider the following linear programming problem :

$$\text{maximize } 3x_1 + 2x_2$$

$$\text{subject to } x_1 + 2x_2 \leq 16, \quad x_1 + x_2 \leq 12, \quad x_1, x_2 \geq 0.$$

The optimal value of this problem is

- (A) 32 (B) 36 (C) 40 (D) none of these.

19. The value of $\int_{-1}^1 (x + x^2 + x^3)e^{-x^2/2} dx$ is

- (A) $2/\sqrt{e}$ (B) $-2/\sqrt{e}$ (C) $2\sqrt{e}$ (D) $-2\sqrt{e}$.

20. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$

- (A) converges absolutely
 (B) converges conditionally and the value is $\log_e 2$
 (C) does not converge
 (D) converges conditionally and the value is $\log_{10} 2$.

21. The solution of the following differential equation

$$(1+x)y \, dx + (1-y)x \, dy = 0$$

- (A) $\log |x| + \log |y| - x + y = \text{constant}$
 (B) $\log |x| - \log |y| - x + y = \text{constant}$
 (C) $\log |x| + \log |y| + x - y = \text{constant}$
 (D) $\log |x| - \log |y| + x - y = \text{constant}$.

Syllabus and Sample Questions for MEII (Economics) 1998-99

Syllabus : MEII

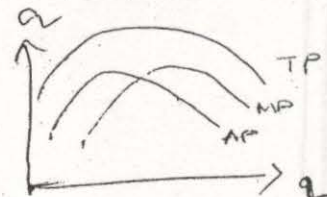
Microeconomics : Consumer theory, Producer theory, Market forms, Welfare economics ; **Macroeconomics :** National income accounting, simple model of income determination and multiplier ; Banking, money supply and inflation. Applicants are expected to have familiarity of these topics at the level of Samuelson and Nordhaus (Economics), and Lipsey (Positive Economics).

Sample Question for MEII (Economics)

1. Prove or disprove the following :

- (a) If over a given range the price-offer curve of a good parallels the axis that represents its quantities, then the demand curve for the good is a straight line in the corresponding range. F
 (b) In a production process, the marginal product must decrease as the level of employment rises whenever marginal product is greater than average product.

[6]



$$M = P_1$$

2. An individual consumes only two commodities, apples and bananas, the quantities of each being denoted by a and b respectively. Suppose he has a utility function.

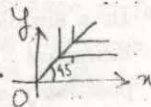
$$W(a, b) = a^2 + b^2, \quad a, b \geq 0$$

- (i) Suppose the price of apples and bananas are Rs. 2 and Re. 1 respectively and that the individual has an income of Rs. 20. At what commodity bundle is his marginal rate of substitution equal to the price ratio? Call this bundle X . Now consider the commodity bundles where the individual spends all his income buying only apples and only bananas. Call these bundles Y and Z respectively. Compare $W(X)$ with $W(Y)$ and $W(Z)$.

- (ii) What is the consumer's optimal commodity bundle (at these prices and income levels)?

3. Two individuals A and B have between them 1 kg. of rice and 1 kg. of dal. They have identical preferences defined by the utility function $U(x, y) = \min\{x, y\}$ where x and y are the amounts of rice and dal respectively consumed by the individual.

- (a) Draw the indifference map of either individual.



- (b) Do you find the following allocations efficient?

(i) A gets 0.3 kg of rice and 0.4 Kg. of wheat; B gets the remainder. NO

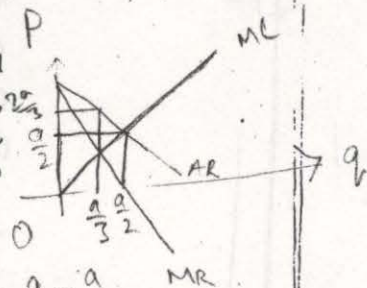
(ii) A gets 0.6 Kg of both rice and wheat; B gets the remainder. Yes.

- (c) If you find any one or both of the allocations to be inefficient, suggest an unequivocal improvement. A gives 0.1 kg wheat & receives 0.1 kg rice.

4. A consumer with a utility function $u = xy$ has the budget constraint $2x + 3y = 36$. However, the consumer is not allowed to consume more than 10 units of x , although he is free to choose any amount of y . Determine his utility maximising (x, y) . $y = 6; x = 9$.

$$\frac{y}{x} = \frac{2}{3} \Rightarrow \frac{3y}{y} = \frac{2x}{3} \Rightarrow 3y = 2x$$

5. An industry demand equation is given by $P = a - Q$ and the marginal cost curve is represented by the equation $MC = Q$ where P is price, Q is quantity and MC is marginal cost. If the industry is served by a monopolist as compared to perfectly competitive firms with the same marginal cost functions, calculate the deadweight loss.



$$\begin{aligned} & \frac{a}{2} \times \frac{a}{6} - \frac{a}{3} \times \frac{a}{6} \\ \Rightarrow & \frac{a^2}{12} - \frac{a^2}{18} \\ \Rightarrow & \boxed{\frac{a^2}{36}} \end{aligned}$$

[7]

$$sdy + yds =$$

$$sdy + yds - bdy$$

$$bdy = sdy + yds$$

$$C + S = Y = C + I$$

$$S = I$$

6. Consider a simple Keynesian model with an investment function $I_0 = I_0 + by$ and a savings function $S = sy$, where $1 > s > b > 0$. By how much does total saving change if there is a small change in marginal propensity to save? bdy

$$sy = I_0 + by$$

$$I_0 = (s-b)y$$

7. A good is produced using only one variable input according to the production function $q = \sqrt{x}$. There are 100 perfectly competitive firms in the market. Derive the supply curve of the industry and discuss the nature of its slope in each of the following cases :

(a) The industry faces a horizontal supply curve of the input at price Rs. 10.00.

(b) The industry faces a vertical supply curve of the input at the quantity of 100 units.

8. Instruction for question numbers 8 (a)–8 (h). For each of the following questions four alternative answers are provided. Choose the answer that you consider to be the most appropriate for a question and write it down in your answer book.

(a) In a simple Keynesian model of the macro economy where mps is marginal propensity to save. If private investment depends positively on aggregate income, the multiplier is

(A) less than $1/mps$

(B) greater than $1/mps$

(C) equal to $1/mps$

(D) equal to $1/(1-mps)$

(b) A farmer produces 100 units of rice. He keeps a part of his output for self-consumption and sells the rest in the market to buy vegetables. If the price of rice goes up, with the price of vegetables remaining unchanged, his budget line

(A) shifts out with the same intercept on the rice-axis.

(B) shifts out with the same intercept on the vegetable-axis.

(C) does not change

(D) shifts in with the same intercept on the rice-axis.

(c) A monopolist with cost function $C = 1.1 + 2Q$ faces a market demand function $P = 4 - Q$. Then in equilibrium its output is

- (A) 0
- (B) 1
- (C) 2
- (D) 3

(d) Consider the simple Keynesian macroeconomy: The consumption function is $C = 20 + 0.9 Y$. Also, $I = G = 20$. The equilibrium GDP is

- (A) 400
- (B) 600
- (C) 360
- (D) 540

(e) Suppose there are 100 barbers in the city of Baleshwar each capable of 100 haircuts a day. The demand for haircuts is given by $2000 - 100p$ where p is the price of haircuts. As a result equilibrium market price is Rs. 10 per haircut. Now suppose demand function shifts to $3000 - 100p$ as a result of a change in fashions favouring a bold look. Other things remaining the same, what will happen to the equilibrium price of a haircut in the long run when there is free entry and exit of barbershops?

- (A) The price of a haircut doubles to Rs. 20.
- (B) The price of a haircut increases but may or may not double
- (C) The price of a haircut does not change
- (D) The price of a haircut decreases to Rs. 5.

(f) A perfectly competitive firm in the short-run uses one variable factor, A , and one fixed factor, B , to produce output X . The firm is in shortrun equilibrium. Which one of the following is true?

- (A) marginal cost of X = price of X
- (B) Price of $X = \frac{\text{Price of } A}{\text{marginal product of } A}$
- (C) Price of $X >$ Average variable cost of X
- (D) All of the above.

(g) In a less than full employment economy there are two classes of income earners—rich and poor. The rich have a higher saving propensity than the poor and investment is autonomously given. Then a redistribution of income from the rich to the poor :

- (A) reduces aggregate saving ;
- (B) reduces aggregate saving and increases aggregate income ;
- (C) increases aggregate saving and increases aggregate income ;
- (D) none of the above ;

(h) The Reserve Bank of India has recently reduced the cash reserve ratio of commercial banks. Normally this should

- (A) decrease the supply of money
- (B) increase the supply of money
- (C) have no effect on the supply of money
- (D) reduce the profit of commercial banks.